

Probability & Statistics – The **Backbone** of **AI/ML**

Linear Algebra – The Backbone

TOPIC :

Problem Solving on Binomial & Poisson distribution

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1/10
 ① For what value of a will be the function
 $f(n) = an; n = 1, 2, 3, \dots, n$
 be the pmf of a discrete random var. X ?
 Find the mean and variance of X .
 $a = ?$

Solⁿ Since $f(n)$ will be a pmf.

$$\text{So, } \sum_{n=1}^n f(n) = 1$$

$$\text{or } \sum_{n=1}^n an = 1$$

$$\text{or, } a(1+2+3+\dots+n) = 1$$

$$\text{or } a \cdot \frac{n(n+1)}{2} = 1$$

$$\text{or } a = \frac{2}{n(n+1)} \checkmark$$

$$\text{Thus, } a = \frac{2}{n(n+1)}$$

$$\text{Mean} = \frac{2n+1}{3}$$

$$\text{Var}(X) = \frac{(n-1)(n+2)}{18}$$

now, $E(X) = \sum_{n=1}^n n f(n)$
 $= a \sum_{n=1}^n n^2 = a(1^2 + 2^2 + 3^2 + \dots + n^2) = 1$
 $= \frac{2}{n(n+1)} \cdot \frac{n(n+1)(2n+1)}{6}$
 $= \frac{2n+1}{3}$ [Putting the value of a].

Again, $E(X^2) = a \sum_{n=1}^n n^3 = a(1^3 + 2^3 + 3^3 + \dots + n^3)$
 Meansquare $= \frac{2}{n(n+1)} \left\{ \frac{n(n+1)}{2} \right\}^2 = \frac{n(n+1)}{2}$

$$\text{So, Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{n(n+1)}{2} - \left(\frac{2n+1}{3}\right)^2 = \frac{(n-1)(n+2)}{18}$$

Binomial distribution

✓ $P(X=r) = {}^nC_r p^r q^{(n-r)}$

As, $P(X)$ is the term of binomial expansion of $(p+q)^n$, is called binomial distribution

Note:- (i) Sum of all prob. in distribution
Sum up to 1.

(ii) Probability of success in n trials is p^n -
(iii) Probability of success in at least one trial $P(X \geq 1) = 1 - P(X=0) = (1 - q^n)$

① Find the Binomial distribution of getting a Six in Three tosses of an unbiased dice.

Solⁿ:- Let X be the random variable of getting Six. Then X can be 0, 1, 2, 3

Here, $n = 3$

p = Probability of getting a Six in a toss = $\frac{1}{6}$

q = Probability of not getting a Six in a toss = $1 - \frac{1}{6} = \frac{5}{6}$

So, required binomial distribution

$$P(X=0) = {}^3C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^{3-0} = 1 \times 1 \times \frac{125}{216} = \frac{125}{216}$$

$$P(X=1) = {}^3C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^{3-1} = 3 \times \frac{1}{6} \times \frac{25}{36} = \frac{75}{216}$$

$$P(X=2) = {}^3C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{3-2} = 3 \times \frac{1}{36} \times \frac{5}{6} = \frac{15}{216}$$

$$P(X=3) = {}^3C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{3-3} = 1 \times \frac{1}{216} \times 1 = \frac{1}{216}$$

X	0	1	2	3
$P(X)$	$\frac{125}{216}$	$\frac{25}{72}$	$\frac{5}{72}$	$\frac{1}{216}$

② Find the probability of getting at least 5 times head-on tossing an unbiased coin for 6 times by using the binomial distribution.

Solⁿ:-

$$p = P(\text{getting on head in a single toss}) = \frac{1}{2}$$

$$q = P(\text{not getting " " " "}) = \frac{1}{2}$$

$$\begin{aligned} P(X \geq 5) &= P(X=5) + P(X=6) \\ &= {}^6C_5 \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^{6-5} + {}^6C_6 \cdot \left(\frac{1}{2}\right)^6 \cdot \left(\frac{1}{2}\right)^{6-6} \\ &= 6 \times \left(\frac{1}{2}\right)^6 + 1 \times \left(\frac{1}{2}\right)^6 \\ &= 6 \times \frac{1}{64} + 1 \times \frac{1}{64} \\ &= \frac{3}{32} + \frac{1}{64} \\ &= \frac{6+1}{64} = \left(\frac{7}{64}\right) \end{aligned}$$

③ There are four fused bulbs in a lot of 10 good bulbs.
 If three bulbs are drawn at random with replacement, find the probability of binomial distribution of the number of fused bulbs drawn.

Solⁿ: $p = P(\text{drawing a fused bulb}) = \frac{4}{10+4} = \frac{2}{7}$
 $q = P(\text{drawing a bulb which is not fused}) = 1 - \frac{2}{7} = \frac{5}{7}$
 $X = \text{event of drawing a fused bulb.}$
 $X = 0, 1, 2, 3$

$$P(X=0) = {}^3C_0 \cdot \left(\frac{2}{7}\right)^0 \cdot \left(\frac{5}{7}\right)^{3-0} = \frac{125}{343}$$

$$P(X=1) = {}^3C_1 \cdot \left(\frac{2}{7}\right)^1 \cdot \left(\frac{5}{7}\right)^{3-1} = \frac{150}{343}$$

$$P(X=2) = {}^3C_2 \cdot \left(\frac{2}{7}\right)^2 \cdot \left(\frac{5}{7}\right)^{3-2} = \frac{60}{343}$$

$$P(X=3) = {}^3C_3 \cdot \left(\frac{2}{7}\right)^3 \cdot \left(\frac{5}{7}\right)^{3-3} = \frac{8}{343}$$

So, required probability distribution:

X	0	1	2	3
P(X)	$\frac{125}{343}$	$\frac{150}{343}$	$\frac{60}{343}$	$\frac{8}{343}$

⑤ Calculate the mean of the Poisson Distribution given that the number of trials is 20 and probability Success is 0.6.

Mean of poisson distribution = np

$$n = 20$$

$$p = 0.6$$

$$np = 20 \times 0.6 = 12$$